Communications

Highly selective hydrohalogenation reaction of substituted 2,3-allenoates

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Abstract The hydrohalogenation reaction of 1-alkyl substituted 1,2-allenyl carboxylic acid esters (**1**) with MX (M = Na, or Li, X = Cl, Br, I) afforded a mixture of β,γ-unsaturated 3-halo-3-alkenoates (**2**) and α,β-unsaturated 3-halo-2-alkenoates (**3**) in HOAc, while using a mixture of HOAc-CF_3CO_2H (1:1) or CF_3CO_2H as the reaction medium the corresponding reaction cleanly produced β,γ-unsaturated 3-halo-3-alkenoates (**2**) as the sole products in high yields. The subsequent coupling reactions were studied.

Keywords Hydrohalogenation, 1,2-allenyl carboxylic acid esters, β,γ-unsaturated 3-halo-3-alkenoates, α,β-unsaturated 3-halo-2-alkenoates, selectivity.

3-Alkenoates are a class of compounds with synthetic importance due to the uniquely located carbon-carbon double bonds. These compounds can be, in principle, prepared *via* either oxidation reactions of the corresponding alcohols/aldehydes or carbon-carbon double bond formation reactions. The major problem for their synthesis is the potential migration of the C=C bond from the β,γ -position to the α,β-position.[1]

Allenes are important intermediates in organic synthesis due to the unique reactivity arising from the existing two orthogonal π-bonds.[2,3] During the course of our study of allene chemistry,[4,5] we found that the hydrohalogenation reaction of unsubstituted 2,3-butadienoic acid and its methyl ester with MX in HOAc afforded 3-halo-3-butenoic acids and esters in high yields and selectivity.[5a] In this paper we wish to report our recent results of the hydrohalogenation reaction of substituted 1,2-allenyl carboxylic acid esters and the corresponding synthetic application of the β,γ-unsatutrated 3-halo-3-alkenoic acid esters thus formed.

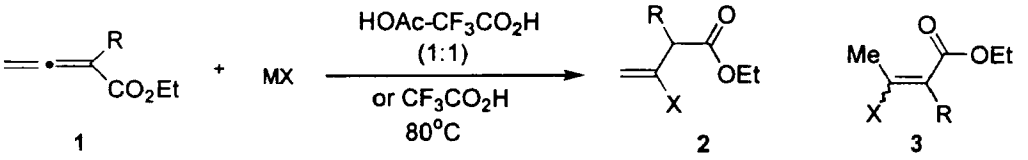
The reaction of 2-alkyl-2,3-alkadienoates with HX is synthetically uninteresting since it afforded a mixture of α,β-unsaturated 2-alkenoates and β,γ-unsaturated 3-alkenoates.[1] Thus, we tried the hydrohalogenation reaction of ethyl 2-methyl-2,3-butadienoate with NaI in HOAc.[4a] However, it also yielded a mixture of α,β-unsaturated 3-iodo-2-methyl-2-butenoate (**3a**) and β,γ-unsaturated 3-iodo-2-methyl-3-

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butenoate (**2a**) (Entries 1 and 2, Table 1). The results with LiBr or LiCl were also poor (Entries 3—5, Table 1). After some trial and error it is interesting to observe that when a 1:1 mixture of CF_3CO_2H and HOAc was used as the reaction medium, the hydrohalogenation reaction went smoothly to afford ethyl 3-halo-2-methyl-3-butenoates (**2**) in fairly good yields and selectivity,[6] the formation of α,β-unsaturated 3-halo-2-methyl-2-butenoates (**3**) were not observed (Entries 6 and 7, Table 1). In addition, the corresponding reaction in pure CF_3CO_2H was much faster and the results were also listed in Table 1 (Entries 8 and 9). Ethyl 2-propyl-2,3-butadienoate reacted similarly with MX in CF_3CO_2H to afford β,γ-unsaturated β-halocarboxylates (**2d**, **2e**, **2f**) in high yields (Entries 10—12, Table 1).

Table 1 Hydrohalogenation reaction of 2-substituted 2,3-allenoates



1 2 3

Entry	MX (equiv.)	R	Solvent	Time (h)	**2** (%)[a]	**3** (%)[a]
1	NaI (1.5)	Me	HOAc	24	**2a** (41)	**3a** (10)
2	NaI (4)	Me	HOAc	72	**2a** (53)[b]	**3a** (14)[b]
3	LiBr-H_2O (1.3)	Me	HOAc	72	**2b**(< 5)	**3b** (< 5)
4	LiBr-H_2O (1.5)[c]	Me	HOAc	120	**2b**(< 5)	**3b** (< 5)
5	LiCl-H_2O (1.8)	Me	HOAc	74	**2c** (~ 10)	**3c** (~ 5)
6	NaI (3)	Me	HOAc-CF_3CO_2H	4	**2a** (95)	*d*
7	LiBr-H_2O (3)	Me	HOAc-CF_3CO_2H	12	**2b** (80)	*d*
8	NaI (2)	Me	CF_3CO_2H	10 min	**2a** (90)	*d*
9	LiBr-H_2O (2)	Me	CF_3CO_2H	40 min	**2b** (88)	*d*
10	NaI (2)	*n*-C_3H_7	CF_3CO_2H	20 min	**2d** (98)	*d*
11	LiBr-H_2O (2)	*n*-C_3H_7	CF_3CO_2H	50 min	**2e** (76)	*d*
12	LiCl-H_2O (3)	*n*-C_3H_7	CF_3CO_2H	10	**2f** (63)	*d*

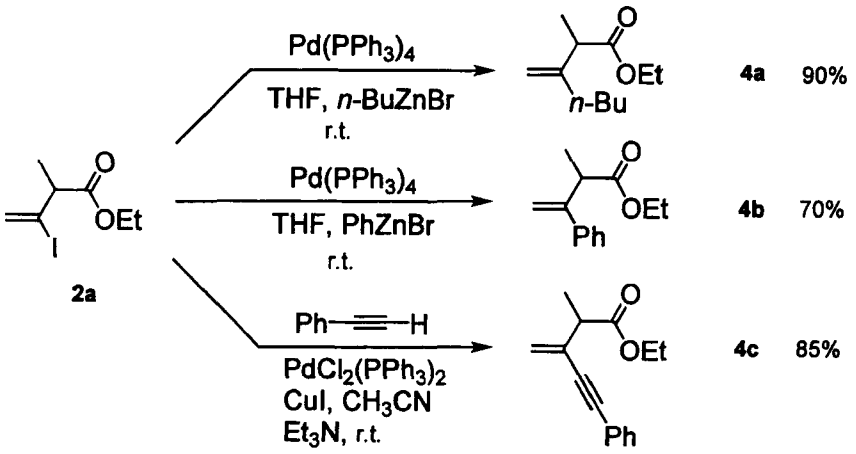
[a] The numbers in parentheses are isolated yields. [b] NMR yield. [c] The reaction was carried out at 40℃. [d] Not observed.

3-Halo-3-alkenoates are expected to be very useful synthetic intermediates, due to the existence of the carbon-halogen bond, the ester group, the carbon-carbon double bond, and the acidic α-proton(s). The carbon-iodine bond in ethyl 3-iodo-2-methyl-3-butenoate (**2a**) was shown to readily cross-couple with terminal alkynes[7,4b] and organozinc reagents[8] under the catalysis of Pd(0) to afford substituted β,γ-unsaturated alkenoates (**4a**—**c**)[9] in high yields. The formation of the corresponding α,β-unsaturated alkenoates was not observed (Scheme 1). Previously, **4b**-type compounds were prepared *via* either the alkylation reaction of β-aryl-β-methyl acrylates[10] or dehydration reaction of β-hydroxycarboxylic acids which could be synthesized with a Reformatsky reaction,[11] while the syntheses of **4a**- and **4c**-type compounds have not

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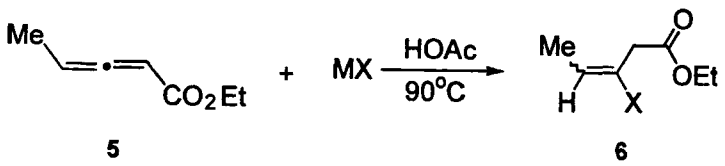
been reported.

Scheme 1



The reaction of ethyl 2,3-pentadienoate (**5**) with NaI, LiBr-H_2O, and LiCl-H_2O in HOAc went smoothly and afforded the corresponding ethyl 3-halo-3-pentenoate (**6**) in high yields (Scheme 2).

Scheme 2

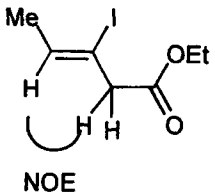


5 **6**

MX	Time (h)	Product	*Z*/*E* ratio	Yield (%)
NaI	6	**6a**	4.7:1	94
LiBr-H_2O	20	**6b**	5.5:1	83
LiCl-H_2O	38	**6c**	1.3:1	70

The configuration of the C=C bond was determined by the 2D NOESY experiment of **6a** (Scheme 3).

Scheme 3

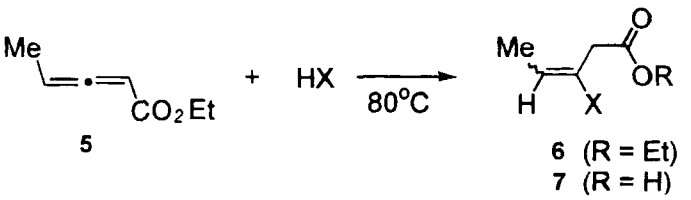


Z-**6a**

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However, the reaction of ethyl 2,3-pentadienoate with HX[12] afforded ethyl 3-halo-3-pentenoates together with 3-halo-3-pentenoic acids (Scheme 4). The hydrolysis of the ester group was observed as the serious side reaction.

Scheme 4



X	Time (h)	Yield of **6** (%)	Yield of **7** (%)
I	4	40	48
Br	24	trace	58
Cl	48	trace	83

In conclusion, we have developed a new general high-yielding reaction for the efficient synthesis of β,γ-unsaturated 3-halo-3-alkenoates. Due to the easy availability of the starting materials[13] and simple procedure, this method will show its utility in organic synthesis.

References

1. Chalchat, J.C.; Duteurtre, P.; Théron, F.; Vessière, R., *C. R. Hebd. Seances Acad. Sc. Ser. C*, **273**, 832 (1971).
2. Schuster, H.F.; Coppola, G.M., *Allenes in Organic Synthesis*, John Wiley & Sons, New York, 1988.
3. *The Chemistry of Ketenes, Allenes, and Related Compounds*, Part 1, Ed. Patai, S., John Wiley & Sons, New York, 1980.
4. a) Ma, S.; Shi, Z., *J. Org. Chem.*, **63**, 6387(1998).
   b) Ma, S.; Shi, Z.; Yu, Z., *Tetrahedron Lett.*, **40**, 2393(1999).
5. a) Ma, S.; Shi, Z.; Li, L., *J. Org. Chem.*, **63**, 4522(1998).
   b) Ma, S.; Wei, Q., *J. Org. Chem.*, **63**, 1026(1999).
   c) Ma, S.; Li, L.; Xie, H., *J. Org. Chem.*, in the press.
6. Typical procedure: To a solution of ethyl 2-methyl-2,3-butadienaote (90.8 mg, 0.72 mmol) in HOAc-CF_3CO_2H (2 mL) was added NaI (325 mg, 2.17 mmol). Then the resulting mixture was heated at 80℃ with stirring. The reaction was monitored by TLC and was complete within 4 h. Then water (1 mL) was added, the acidic solution was neutralized with sat $NaHCO_3$ solution, extracted with ether (4 × 5 mL). The combined organic layer was washed subsequently with 5% $Na_2S_2O_3$ and brine and dried over $MgSO_4$. Filtration and evaporation afforded 173.2 mg (95%) of pure **2a**: ^{1} ν_{max} 1740, 1615 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 6.18(d, J = 1.4 Hz, 1H), 5.82(d, J = 1.4 Hz, 1H), 4.18—4.06(m, 2H), 3.23(q, J = 6.7 Hz, 1H), 1.25—1.16(m, 6H). *m/z*: 254(M^+), 127(100). In other cases further purification could be realized with chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 100/1).
7. Sonogashira, K.; Tohda, Y.; Hagihara, N., *Tetrahedron Lett.*, 4467(1975).

16147065, 1999, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.19990170517 by Zhejiang University, Wiley Online Library on [30/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

8. For a seminal paper, see: Negishi, E.; King, A.O.; Okudado, N., *J. Org. Chem.*, **42**, 1821(1977).

9. **4a**: Liquid: ν_{max}: 1735, 1640 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 4.90(s, 1H), 4.86 (s, 1H), 4.12(q, J = 7.1 Hz, 2H), 3.12(q, J = 7.1 Hz, 1H), 2.04(q, J = 4.3 Hz, 2H), 1.42—1.20 (m, 10H), 0.89(t, J = 7.1 Hz, 3H). m/z: 184(M^+), 69(100). HRMS: $C_{11}H_{20}O_2$. Cacld: 184.1463. Found: 184.1468.

4c: Liquid: ν_{max} 2359, 1735, 1600 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 7.44—7.41 (m, 2H), 7.31—7.28 (m, 3H), 5.57(s, 1H), 5.46(s, 1H), 4.22—4.12(m, 2H), 3.33(q, J = 7.0 Hz, 1H), 1.42(d, J = 7.0 Hz, 3H), 1.27(t, J = 7.0Hz, 3H). m/z:228(M^+), 156(100). HRMS: $C_{15}H_{16}O_2$. Cacld: 228.1150. Found: 228.1145.

10. a) Kano, S.; Yuasa, Y.; Yokomatsu, T.; Asami, K.; Shibuya, S., *J. Chem. Soc., Chem. Commun.*, 1717 (1986).

b) Kano, S.; Yuasa, Y.; Shibuya, S., *Heterocycles*, **27**, 1437(1988).

11. a) Theine, S.A.; Traynham, J.G., *J. Org. Chem.*, **39**, 153(1974).

b) Delaunay, J.; Orliac, A.; Moing, L.; Simonet, J., *Tetrahedron*, **44**, 7089 (1988).

12. For the reaction of 2,3-allenoate with HX, see:

a) Rosenthal, R.W.; Schwartzman, L.H:, *J. Org. Chem.*, **24**, 836(1959).

b) Kurtz, P.; Gold, H.; Disselnkötter, H., *Ann.* **624**, 1(1959).

13. Lang, R.W.; Hansen, H.-J., *Helv. Chim. Acta.*, **63**, 438(1980).

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